

PROBLEM & EXPERIENCE

Nobody ever learned from a quiz

Rehnuma turns any document into a story you watch, then a game you play, in English or Urdu. It works out what you understand from how you talk — and never gives a test.

1 Bring anything

PDF, Word, PowerPoint, spreadsheet, web link or pasted text.

2 Watch the story

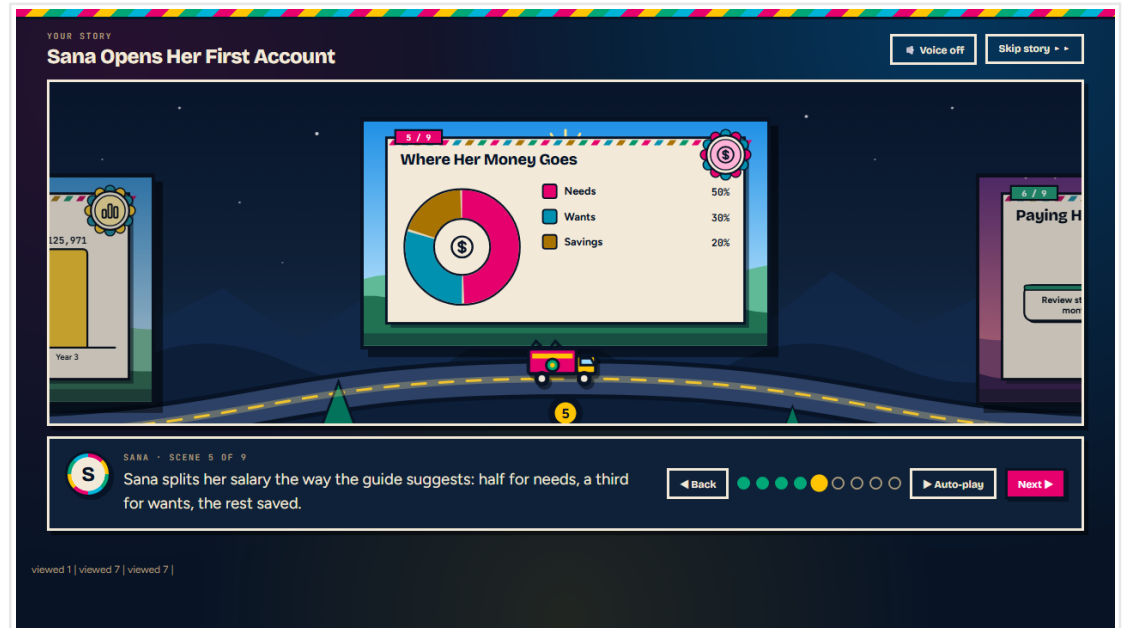
An illustrated road-trip through the ideas, drawn from your own document.

3 Play it

Missions and choices with consequences. Speak or type, Urdu or English.

4 See progress

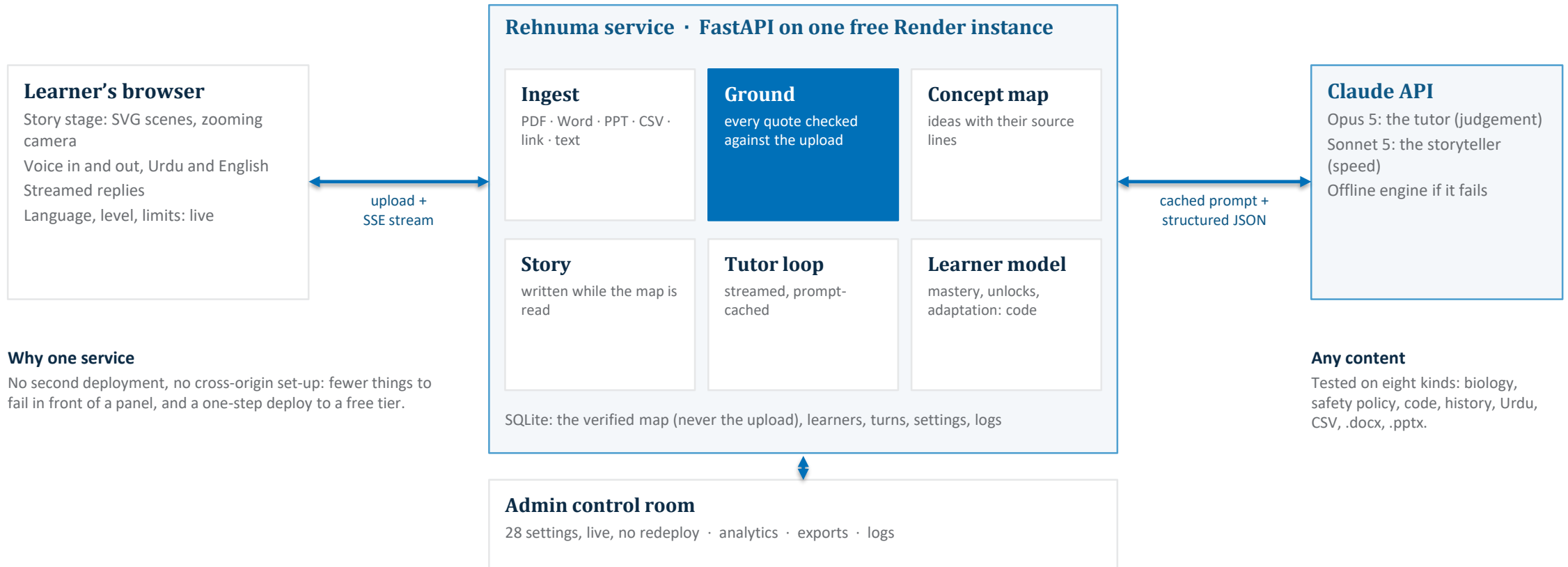
Concepts unlock as understanding grows. No pass mark anywhere.



Above: a story scene, its chart drawn from the document's own figures. Below: the game, and the same in Urdu, right-to-left.

One service, three trust boundaries

It reads a document, proves every idea against it, tells the story, then holds the conversation — and keeps its own arithmetic for what you have understood.



Built for a bank

Two risks matter most: a tutor that invents a fact about a financial product, and a document written to hijack it. Everything else is layered around those.

Invented facts	Every quote is checked against the upload before it is saved; a claim must cite a real quote; off-source questions are refused and flagged. 100% verbatim across all eight content types tested.
Hijacking through content	Uploaded text and chat messages are marked as data and stripped of our prompt delimiters. The story can only name a layout or icon: it never supplies markup.
Attacks on the server	Web fetch refuses private networks and redirects to them; file type, size and archive limits; strict CSP with no inline or third-party script; frame, sniff and referrer protection.
Access and abuse	Admin cookie is signed and expires; no default password in production; per-client rate limits that hold behind the proxy (verified live); a daily spend cap.
Bad settings	Every setting is validated on the server. A rate limit of –5 used to break every request.
Privacy	The upload itself is never stored, only the verified map. No personal data: a first-name label and a random ID. Retention purge, and the learner or an admin can erase a session.
Reliable & observable	Structured logs, /health, an in-app log viewer, per-turn latency and tokens. If the AI service fails, the offline engine takes over and says so.

Fast, adaptive, measured

Streaming and early preparation keep it quick. Adapting on measured evidence, and recording that evidence, is how it learns — and how we know.

FAST AND FRUGAL · measured on the live deployment

~3 s

to the first words of a reply, streamed

≤ 12 s

for the story, written while the map is read

72%

of prompt input served from cache

+14%

average mastery gain in the demo sessions

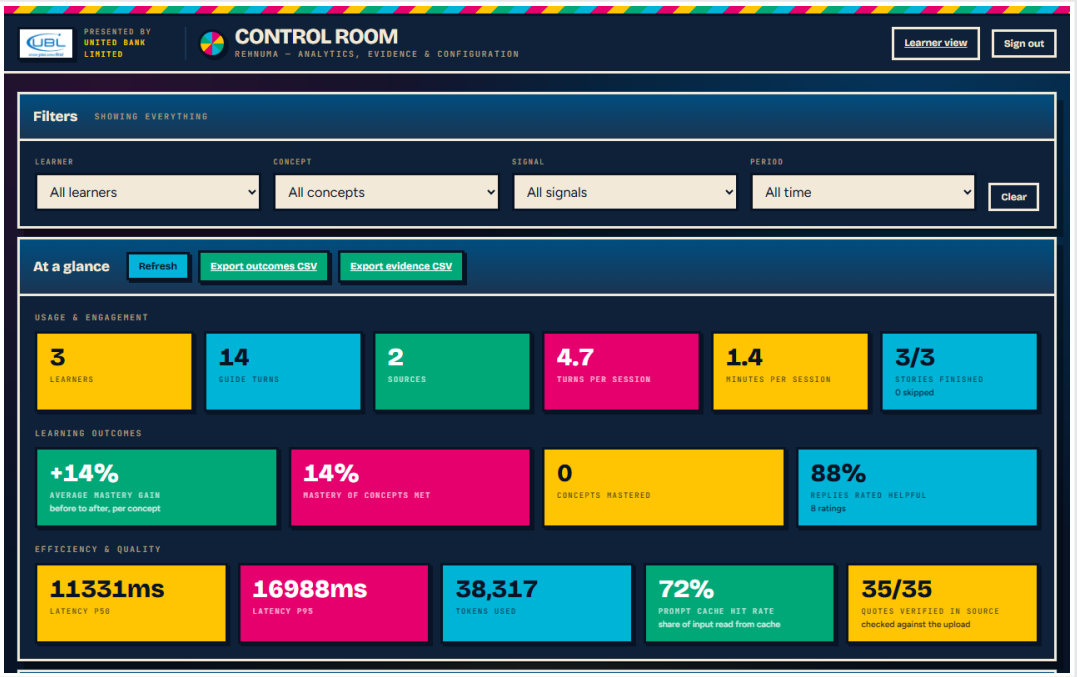
ADAPTS FROM MEASURED EVIDENCE, NOT THE MODEL’S MOOD

Difficulty, support, pace and the next concept are set in code from hints, streaks and strong answers.

Six behaviours judged every reply, each kept beside the learner’s own words: applied it · corrected themselves · used the vocabulary · asked ahead · leaned on a hint · remembered.

Language, level or a limit like “five minutes, on a phone” can change mid-session; the next reply obeys.

Hosting is \$0 on Render’s free tier. Honest limits: an idle instance takes ~40 s to wake and free storage resets on redeploy.



The control room, read from three real learner sessions (English, Urdu, mixed): +14% mastery gain, 88% of 8 replies rated helpful, 72% cache hits, 35 of 35 quotes verified.

WHAT MANAGEMENT SEES

Mastery gain before → after · engagement and story completion · learning curve · where learners struggle · helpfulness · latency · cache hits. Filter by learner, concept, signal or period; export CSV; print a one-page learner report.

How every requirement is covered

The twelve things the brief asks a solution to demonstrate: what covers each one here, and what powers it.

REQUIREMENT	HOW IT IS COVERED HERE	TECHNOLOGY
Any content → experience	Any document becomes a verified concept map, an illustrated story, then a game — not a summary or a quiz.	PyMuPDF · Word/PowerPoint parsing · CSV→prose · SSRF-guarded fetch · Claude structured outputs
Adaptive personalisation	Difficulty, support, pace and next concept come from measured hints and answers; language, level and limits change live.	Deterministic Python learner model · per-learner overrides in SQLite
Gamification & interaction	Story road with a truck, missions, puzzles, boss, choices with consequences, XP, streaks, badges, unlocks.	Vanilla-JS SVG scene engine · state machine in code
Assessment without tests	Six behaviours judged each reply with the learner’s own words as evidence; before/after mastery per concept.	Claude structured JSON (signals + deltas) · clamped arithmetic in Python
Voice + accessibility	Urdu/English speech in and out, code-switching, right-to-left Nastaliq, talk mode, text fallback, reduced motion, ARIA.	Web Speech API · Noto Nastaliq Urdu · CSS/ARIA
Accuracy & grounding	Every quote is checked against the upload; claims cite a real quote; off-source questions are refused and flagged.	grounding.py (normalised match + difflib repair) · citation chips
Highly configurable	28 settings — level, language, tone, mechanics, thresholds, custom rules — change live, validated.	Config service on SQLite · admin control room · pydantic
Dashboards & reports	Learner, engagement, mastery, usage and outcome views; filters; CSV exports; printable learner report.	Chart.js (self-hosted) · FastAPI analytics · CSV/HTML export
Efficient & effective	First words in ~3 s; story prepared while the map is read; 72% prompt-cache hits; measurable mastery gain.	SSE streaming · prompt caching · background threads · Opus/Sonnet right-sizing
Free cloud deployment	One stateless service on a free tier, redeployed from GitHub on every push; scales sideways.	Render free tier · render.yaml · GitHub · uvicorn/FastAPI
Cybersecurity & privacy	Signed admin sessions, rate limits, strict CSP, validated inputs, SSRF guard, upload never stored, erase and retention.	HMAC cookies · slowapi · CSP headers · pydantic · SQLite purge
Reliable & observable	JSON logs, /health, in-app event log, per-turn latency and tokens; offline engine takes over if the AI service fails.	Python logging · events table · /health · automatic fallback engine